

# Mohan Konugolu

## Robotics Engineer

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## SUMMARY

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Robotics engineer with 5+ years across robotics consulting, autonomous driving, and mobile-robot startups. Proven from on-site deployment and perception through to full autonomy stacks and reinforcement learning, with deep NVIDIA Isaac Sim, Isaac Lab, and ROS / ROS 2 expertise. Top Rated freelance consultant building digital twins and RL pipelines for robotics companies worldwide, with every project containerized in Docker for reproducible deployment.

## TECHNICAL SKILLS

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**Languages:** Python, C++, Bash

**Robotics & Middleware:** ROS, ROS 2, Nav2, real-time control

**Simulation & Digital Twins:** NVIDIA Isaac Sim, Isaac Lab, Isaac Gym, Omniverse USD, Gazebo

**Reinforcement & Imitation Learning:** PPO, SKRL, Isaac Lab, reward shaping, sim-to-real, motion retargeting (SMPL-X, PHC), whole-body control (HOVER)

**Perception & Computer Vision:** OpenCV, YOLO, MiDaS, point-cloud processing, 6-DoF pose estimation (FoundationPose), Isaac ROS

**Deep Learning:** PyTorch, CNNs (MobileNetV3), time-series classification (MiniRocket)

**Motion Planning & Control:** pure pursuit, Timed Elastic Band (TEB), clothoid path generation, path tracking, LiDAR safety zones

**Mapping & Localization:** 2D-LiDAR mapping (SLAM), localization, autonomous navigation and docking

**Sensors & Hardware:** 2D/3D LiDAR, solid-state LiDAR, IMU, RGB-D cameras, force/torque sensors; AMRs, manipulators, humanoids (Unitree G1)

**Deployment & Tools:** Docker (all projects containerized), Git, Linux

## PROFESSIONAL EXPERIENCE

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### Robotics Consultant (Freelance) — Upwork Feb 2024 – Present

*Remote*

- Top Rated robotics consultant (100% Job Success, 20 jobs, 597+ hours) delivering NVIDIA Isaac Sim simulation and reinforcement-learning solutions to robotics companies worldwide; all projects containerized with Docker for easy deployment.
- Ported complete robot software stacks to ROS 2 / Isaac Sim and implemented a FoundationPose 6-DoF object-pose pipeline using NVIDIA Isaac ROS.

#### Robot Factory Digital Twin (AMRs & Manipulators)

- Built large-scale Isaac Sim factory environments with multiple coordinated robots and simulated LiDAR, camera, and IMU streams.
- Wired the twin to ROS 2 and Isaac Lab so client teams could validate control logic and train RL policies before hardware rollout.

#### Custom Humanoid Simulation & RL Locomotion (Hyperspawn)

- Imported an open-source humanoid with 4-bar linkages and a Stewart platform into Isaac Sim and solved the closed-chain kinematics so the simulation matched the real robot.
- Trained a bipedal walking policy with reinforcement learning and integrated an SMPL-X motion-retargeting (PHC) pipeline for imitation + RL; credited as an open-source contributor.
- Built the RL training pipeline in Isaac Lab and validated stable, real-robot-like locomotion in simulation.

#### Humanoid Whole-Body Gesture Control (Unitree G1)

- Trained the Unitree G1 to perform naturalistic conversational gestures using the HOVER whole-body-control framework.
- Adapted the SMPL-X motion-retargeting pipeline and extended the task to five-fingered hands for lifelike hand movement while talking.

#### **Dexterous Brick-Laying Manipulation**

- Built a custom Isaac Sim environment with a five-fingered arm, fingertip force sensors, and IMUs.
- Trained with PPO (SKRL) to approach, grasp with all five fingers, lift, and place bricks at a target location.
- Tuned the reward function for a smooth approach-grasp-lift-place sequence and reliable placement accuracy.

#### **RL Motion Control for Autonomous Navigation**

- Built a custom-vehicle Isaac Sim scene (LiDAR, camera, IMU) bridged to ROS 2 and integrated an RL policy with Nav2 for smooth low-speed navigation and pedestrian avoidance.
- Added custom low-speed smoothing for stable steering and validated point-to-point navigation around slow-moving pedestrians.

### **Software Developer — Bosch Global Software Technologies (BGSW)** Sep 2022 – Feb 2024

*India*

- Developed machine learning for autonomous driving with a focus on computer-vision depth perception and reinforcement-learning vehicle control.

#### **Monocular Depth Estimation**

- Trained a deep network (YOLO boxes + MiDaS relative depth) for absolute object-level depth — 0.18 m MAE in-house, 1.1 m on the nuScenes dataset.
- Selected for a poster presentation at Bosch Germany's internal AAI research conference.

#### **LiDAR Cover-Glass Blockage Detection**

- Detected cover-glass occlusion on a solid-state LiDAR with MiniRocket (photon time-series) and MobileNetV3 (3-channel point-cloud images), both above 90% accuracy.
- Compared the raw-photon time-series and point-cloud-image approaches and selected the stronger model for deployment.

### **Robotics Engineer — ZeroPoint Robotics** Aug 2021 – Aug 2022

*India*

- Core software engineer for an autonomous material-handling mobile robot; owned the full stack from safety to motion planning.

#### **Autonomous Mobile Robot Software Stack**

- Developed a double-look-ahead pure-pursuit path tracker and clothoid-based continuous-curvature path generation; deployed industry-standard Timed Elastic Band (TEB) for efficient local planning.
- Coded 3-zone 2D-LiDAR safety monitoring and a trolley-leg detection algorithm for autonomous docking.

### **Deployment Engineer — THRSL** Dec 2020 – Jul 2021

*India*

- Developed pallet-detection perception and executed on-site deployment of the mobile robot.

#### **Autonomous Pallet-Jack Commissioning**

- Installed two autonomous battery-operated pallet jacks in a working factory, setting up 2D-LiDAR mapping, localization, route verification, and motion-controller tuning.
- Tuned route-following and docking behavior and coordinated on-site acceptance testing with the factory team.
- Validated safety sensors and functions and ran extended long-term operation across predetermined routes and missions.

## **EDUCATION**

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### **MSc, Robotics and Artificial Intelligence** Sep 2024 – Sep 2025

*Cardiff Metropolitan University — Cardiff, UK*

### **BEng, Mechatronics** 2016 – 2020

*Chandigarh University — Punjab, India*